PROGRAM-8

1. Problem Statement

Develop a manufacturing cost and profitability analysis system that calculates the total production cost by considering raw material costs, labour charges, material wastage, and machine operating costs. The system should determine the selling price, calculate the expected profit, and identify whether the production meets a specified profit target.

2. Objectives

1. **To calculate the total cost of raw materials used in production.**
2. **To calculate labour charges based on working hours and labour rates.**
3. **To estimate additional costs caused by material wastage.**
4. **To calculate machine operating and maintenance costs.**
5. **To determine the total manufacturing cost and selling price.**
6. **To calculate the expected profit and profit percentage.**
7. **To check whether the production meets the specified profit target.**

3. Key Features

- **Material Cost Calculation: Calculate the cost of raw materials required for manufacturing products.**
- **Labour Cost Calculation: Determine labour charges based on working hours and wages.**
- **Wastage Cost Calculation: Estimate the additional material cost due to wastage during production.**
- **Machine Cost Calculation: Include machine operating and maintenance expenses.**
- **Total Production Cost: Calculate the combined cost of all manufacturing activities.**
- **Selling Price Calculation: Determine the selling price of the manufactured products.**
- **Profit Analysis: Calculate the profit or loss for each production batch.**
- **Profit Target Verification: Check whether the expected profit reaches the specified target.**
- **Cost Summary Report: Display material, labour, wastage, machine, and total production costs.**

4. Applications

1. **Manufacturing industries.**
2. **Small-scale production units.**
3. **Automobile manufacturing companies.**
4. **Textile and garment industries.**
5. **Electronics manufacturing units.**
6. **Furniture manufacturing businesses.**
7. **Food processing industries.**

5. Example Calculation

Suppose a company manufactures 100 products with the following expenses:

Cost Component	Amount
Raw Material Cost	₹5,000
Labour Charges	₹2,000
Wastage Cost	₹500
Machine Cost	₹1,000
Total Manufacturing Cost	₹8,500

Assume the following details:

- **Number of products: 100**
- **Selling price per product: ₹100**
- **Target profit: ₹2,000**

Calculations

Step 1: Total Manufacturing Cost

Total Cost = Material Cost + Labour Cost + Wastage Cost + Machine Cost

Total Cost = ₹5,000 + ₹2,000 + ₹500 + ₹1,000

Total Manufacturing Cost = ₹8,500

Step 2: Total Revenue

Total Revenue = Number of Products × Selling Price per Product

Total Revenue = 100 × ₹100

Total Revenue = ₹10,000

Step 3: Total Profit

Profit = Total Revenue − Total Manufacturing Cost
Profit = ₹10,000 − ₹8,500
Total Profit = ₹1,500
Step 4: Profit Target Verification

- **Target Profit = ₹2,000**
- **Actual Profit = ₹1,500**

Result: The production does not meet the specified profit target.
The company earns a profit of ₹1,500, which is ₹500 below its target.
6. Technologies Used

- **Programming Language: Python**
- **User Interface: HTML, CSS, and JavaScript (optional)**
- **Database: MySQL or SQLite (optional)**

7. Expected Outcome
The system will help manufacturers understand their production expenses, calculate selling prices, estimate profits, and verify whether production meets the desired profit target. It can support better cost control, reduce wastage, and improve business profitability.

Programm

```
# Manufacturing Cost and Profitability Analysis

# Input details
product_name = input("Enter product name: ")
quantity = int(input("Enter quantity produced: "))

material_cost = float(input("Enter material cost per unit (Rs.): "))
labour_cost = float(input("Enter labour cost per unit (Rs.): "))
machine_cost = float(input("Enter machine cost per unit (Rs.): "))
wastage_percentage = float(input("Enter wastage percentage: "))
selling_price = float(input("Enter selling price per unit (Rs.): "))
profit_target = float(input("Enter target profit percentage: "))

# Calculate material cost including wastage
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```
total_material_cost = material_cost * quantity
wastage_cost = total_material_cost * wastage_percentage / 100

# Calculate other production costs
total_labour_cost = labour_cost * quantity
total_machine_cost = machine_cost * quantity

# Calculate total production cost
total_cost = (
    total_material_cost
    + wastage_cost
    + total_labour_cost
    + total_machine_cost
)

# Calculate selling revenue and profit
total_revenue = selling_price * quantity
profit = total_revenue - total_cost

# Calculate profit percentage
if total_cost > 0:
    profit_percentage = (profit / total_cost) * 100
else:
    profit_percentage = 0

# Check whether production meets profit target
print("\n======= MANUFACTURING REPORT =======")
print("Product Name:", product_name)
print("Quantity Produced:", quantity)
print("Total Material Cost: Rs.", round(total_material_cost, 2))
print("Wastage Cost: Rs.", round(wastage_cost, 2))
print("Total Labour Cost: Rs.", round(total_labour_cost, 2))
print("Total Machine Cost: Rs.", round(total_machine_cost, 2))
print("Total Production Cost: Rs.", round(total_cost, 2))
print("Total Sales Revenue: Rs.", round(total_revenue, 2))
print("Total Profit/Loss: Rs.", round(profit, 2))
print("Profit Percentage:", round(profit_percentage, 2), "%")
```

```
if profit_percentage >= profit_target:
    print("Result: Production meets the profit target.")
else:
    print("Result: Production does not meet the profit target.")

print("====================================")
```

OUTPUT:

```
Enter product name: ponse dream flavor
Enter quantity produced: 800
Enter material cost per unit (Rs.): 1000
Enter labour cost per unit (Rs.): 32
Enter machine cost per unit (Rs.): 10
Enter wastage percentage: 78
Enter selling price per unit (Rs.): 45
Enter target profit percentage: 98

======= MANUFACTURING REPORT =======
Product Name: ponse dream flavor
Quantity Produced: 800
Total Material Cost: Rs. 800000.0
Wastage Cost: Rs. 624000.0
Total Labour Cost: Rs. 25600.0
Total Machine Cost: Rs. 8000.0
Total Production Cost: Rs. 1457600.0
Total Sales Revenue: Rs. 36000.0
Total Profit/Loss: Rs. -1421600.0
Profit Percentage: -97.53 %
Result: Production does not meet the profit target.
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Process finished with exit code 0
```